

Performance Test Report — AI Log Mining

Run ID: 5b994405 | Generated: 2026-07-01 09:50 UTC | Performance Engineering

Verdict: **FAIL**

Test Summary

Total findings	SLA breaches	Anomalies	Trends	Patterns
673	11	652	5	5

LoadRunner Transaction Results

Transaction	Avg(s)	Min	P90	Max	Passed	Failed	TPS	SLA
Login	0.45	0.38	0.8	1.8	None	0	200.0	INFO
Search Products	0.3	0.25	0.62	0.9	None	0	350.0	INFO
Order Confirm	0.35	0.28	0.7	1.5	None	0	120.0	INFO
Add to Cart	0.55	0.44	0.98	2.4	None	3	180.0	INFO
Checkout	1.2	0.95	2.1	6.8	None	12	80.0	INFO
Payment	0.95	0.8	1.75	5.2	None	8	60.0	INFO

Server Resource Utilization

SQL Metrics (SQL)

SQL Server counters

Buffer cache hit	82.24%
Page life exp.	50.0s
Blocked procs	20
Deadlocks/s	1.07
Lock waits/s	24.95
Recompiles/s	34.17

[!] Buffer cache hit 82.24% — below 95.0% minimum

[!] Page life expectancy 50.0s — near 300s floor

[!] Blocked processes: 20 — exceeded 5 threshold

SERVER01 (PerfMon)

PerfMon — system

CPU (peak)	97.93%
Memory %	98.0%
Disk queue	4.49
Network MB/s	49.9

[!] CPU peaked at 97.93% — exceeded 85% threshold

[!] Memory at 98.0% — near threshold

[!] Disk queue length 4.49 — exceeds normal threshold

IIS (IIS)

IIS — web traffic

Total requests	500
Error rate	5.8%
Avg resp (ms)	350
P95 resp (ms)	576
P99 resp (ms)	588
HTTP 500s	7
HTTP 4xxs	22
Avg req/sec	0.6
Peak TPS (10s)	8

Endpoint Performance (top 15 by P95)

HTTP Status Distribution

[!] 7 HTTP 5xx errors — 1.4% of requests

[ok] Peak load: 8 req/10s at 2024-06-19 09:07:00

Executive Summary

The load test revealed significant performance degradation primarily due to memory pressure, which led to a cascade of failures. Memory availability decreased sharply, triggering excessive garbage collection and CPU saturation. This resulted in increased disk queue lengths and prolonged response times, causing HTTP 5xx errors and transaction failures. The probable root cause is inefficient memory usage or a memory leak. Immediate remediation should focus on optimizing memory allocation and garbage collection settings to alleviate pressure on system resources.

Cause → Effect Chain

- 1. Memory pressure leads to increased garbage collection activity.
- 2. Increased garbage collection causes CPU saturation as observed with 97.9278% CPU usage.
- 3. CPU saturation results in slower processing times, contributing to increased disk queue lengths.
- 4. Disk queue saturation causes delays in I/O operations, impacting application response times.
- 5. Prolonged response times lead to HTTP 5xx errors and transaction failures.

Root Cause Hypotheses

Hypothesis	Confidence	Evidence
Application misconfiguration is leading to systemic failures.	Medium	Consistent transaction failures and HTTP 5xx errors suggest systemic issues.
Memory leak or inefficient memory usage is causing memory pressure.	High	Decreasing memory_mem_available_mb and high mem_committed_pct (98%) indicate memory pressure.
Excessive garbage collection is leading to CPU saturation.	High	High CPU usage (97.9278%) coincides with memory pressure and decreasing memory_mem_available_mb.
I/O operations are bottlenecked due to disk queue saturation.	Medium	Disk queue length increasing and exceeding threshold (4.4871 actual vs 2.0 threshold).
Database contention is causing transaction failures.	Medium	SQL deadlocks and lock waits exceed thresholds, impacting database performance.

SLA & Threshold Breaches

Metric	A	T	Severity
cpu_pct			WARN

processor_cpu_pct	WARN
mem_committed_pct	WARN
disk_queue_len	CRITICAL
http_status	CRITICAL
sql_buffer_cache_hit_pct	WARN
sql_page_life_expectancy_s	CRITICAL
sql_deadlocks_sec	CRITICAL
sql_lock_waits_sec	CRITICAL
sql_processes_blocked	CRITICAL
sql_recompilations_sec	CRITICAL

Event Timeline (top 25)

Timestamp	Source	Metric	Severity
2024-06-19 08:00:06	loadrunner	sla_breach	critical
2024-06-19 08:00:10	blg	memory_mem_available_mb	warn
2024-06-19 08:00:10	blg	memory_mem_committed_pct	critical
2024-06-19 08:00:10	blg	memory_mem_committed_pct	warn
2024-06-19 08:00:10	blg	process_proc_handle_count	warn
2024-06-19 08:00:10	blg	process_proc_working_set_b	warn
2024-06-19 08:00:10	sql	deadlock	critical
2024-06-19 08:00:20	blg	memory_mem_available_mb	warn
2024-06-19 08:00:20	blg	memory_mem_committed_pct	critical
2024-06-19 08:00:20	blg	memory_mem_committed_pct	warn
2024-06-19 08:00:20	blg	network_interface_net_bytes_total_sec	warn

2024-06-19 08:00:20	blg	process_proc_handle_count	warn
2024-06-19 08:00:20	blg	process_proc_working_set_b	warn
2024-06-19 08:00:30	blg	memory_mem_available_mb	warn
2024-06-19 08:00:30	blg	memory_mem_committed_pct	critical
2024-06-19 08:00:30	blg	memory_mem_committed_pct	warn
2024-06-19 08:00:30	blg	process_proc_handle_count	warn
2024-06-19 08:00:30	blg	process_proc_working_set_b	warn
2024-06-19 08:00:40	blg	memory_mem_available_mb	warn
2024-06-19 08:00:40	blg	memory_mem_committed_pct	critical
2024-06-19 08:00:40	blg	memory_mem_committed_pct	warn
2024-06-19 08:00:40	blg	process_proc_handle_count	warn
2024-06-19 08:00:40	blg	process_proc_working_set_b	warn
2024-06-19 08:00:50	blg	memory_mem_available_mb	warn
2024-06-19 08:00:50	blg	memory_mem_committed_pct	critical